CARELOOP

INTELLIGENT CARE COORDINATION & WELLNESS COMPANION

From fragmented healthcare journeys to connected, intelligent care.

AI-POWERED CARE COORDINATION & WELLNESS PLATFORM | GNANI AI INTERNSHIP PROJECT

THE PROBLEM

Healthcare journeys fragment across appointments, instructions, follow-ups and caregiver communication. Families struggle to know **what happens next**, **what is pending**, **who is responsible** and which updates are urgent, leading to missed follow-ups, unclear ownership and disconnected care.

OUR SOLUTION

CareLoop brings patients, families and caregivers into one workflow: structured care plans, milestone tracking, overdue and gap alerts, caregiver ownership and handoffs, searchable notes, and natural-language or spoken requests turned into reviewable actions, with people in control of care decisions.

> ***"CareLoop does not simply store healthcare information; it connects information, responsibilities, actions, and follow-ups into a coordinated care workflow."***

SIX CORE CAPABILITIES

01 — CARE PLAN
Turns care requirements into prioritized plans with checklists, recurring tasks, task dependencies, progress notes and full activity history.

02 — CARE JOURNEYS
Tracks end-to-end journeys with milestones, pending activities and progress, and surfaces blockers and journeys past their target date.

03 — SENTINEL
Rule-based scan that flags overdue, blocked, unowned or unacknowledged tasks and possible gaps, creating alerts for human review.

04 — TEAM HANDOFFS
Supports caregiver task ownership and workload visibility; tracks handoffs through acceptance with an event history and 24h follow-up alert.

05 — CARE CONTEXT
Searchable care notes linked to tasks and journeys, plus a concise Visit Brief that gathers open actions, updates and recent completions.

06 — CAREPILOT
Interprets typed or transcribed care requests (English/Hindi) into editable, structured proposals that are saved only after human confirmation.

GNANI AI - VOICE INTELLIGENCE

Speech-to-Text: IMPLEMENTED
Gnani Prisma v3 REST (`POST /stt/v3`) transcribes uploaded audio (WAV, MP3, FLAC, OGG, M4A, AAC; up to 30 s) in multilingual, code-mixed mode. The transcript fills CarePilot's request box for review. Includes validation, provider error mapping and timeouts; the key stays server-side. Offline mock test included; 'connected' is shown only after a live success.

Text-to-Speech: PLANNED
Spoken responses are not in the current repository. Planned as the next step: Gnani TTS to read back confirmations and briefs for hands-free, accessible use. Voice complements all six capabilities, with CarePilot as the entry point today.

Audio input ▶ **Gnani STT** ▶ **Transcript** ▶ **CarePilot logic** ▶ **Editable proposal** ▶ **Human confirms**

Planned extension: CarePilot response > Gnani TTS > spoken reply

TECHNOLOGY & ARCHITECTURE

Python · **Flask** · **Jinja/HTML templates** · **SQLite** · **python-dotenv** · **requests** · **Gnani Prisma v3 STT**
Rule-based monitoring · **Local intent engine** · **Git & GitHub** · **venv** · **VS Code**

User interaction ▶ **CareLoop Flask web app** ▶ **Six core capabilities** ▶ **Application logic + SQLite** ▶ **Gnani STT integration** ▶ **Structured, reviewable output**

The Flask app coordinates the modules; Gnani provides speech-to-text. Monitoring and request interpretation are **rule-based Python logic** (not machine learning or generative AI).

USERS, IMPACT & VISION

TARGET USERS

Patients, family members, caregivers, and individuals coordinating ongoing care activities.

EXPECTED IMPACT

Improves continuity of care, reduces overlooked follow-ups, strengthens caregiver accountability, and makes care information and responsibilities easier to see and access.

LONG-TERM VISION

A proactive, connected, human-centered care ecosystem with accessible voice interaction. CareLoop organizes care; it does not diagnose or decide treatment.

GITHUB REPOSITORY

https://github.com/anjali-r05/CareLoop

LIVE DEMO (VIDEO WALKTHROUGH)

https://youtu.be/wleIqsLcuDE

RESPONSIBLE AI & SAFETY

API credentials are kept in environment variables (a local .env file, not in the repository). Sensitive information is handled responsibly. AI-generated output, including transcripts, is never presented as a verified clinical decision: every proposal is reviewed and confirmed by a person, and important care decisions remain under appropriate human supervision.

CareLoop supports care organization and coordination; it is not a substitute for professional medical advice, diagnosis, treatment, or emergency care.